PRODUCT CARD

ABS V0

1. GENERAL INFORMATION ABOUT THE PRODUCT

A specialized ABS variant characterized by a very high degree of flame retardancy, meeting UL94 V0 standards even under the most stringent conditions, while maintaining the mechanical parameters of ABS.

Main ABS V0 features:

- meets the V0 flammability criteria according to UL-94, even for 1,5 mm details,
- good mechanical strength, rigidity, and hardness,
- high impact resistance,
- good thermal stability,
- permissible continuous operating temperature above 70°C,
- good electrical insulation properties.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ISO			
PHYSICAL				
Density	ASTM D792	-	g/cm^3	1.16
Water absorption to saturation	ASTM D570	23°C/sat.	%	0.2
Processing shrinkage II/⊥		-	%	0.1~0.3
MECHANICAL				
Yield strength	527-1,-2	50mm/min	MPa	39
Elongation at break	527-1,-2	50mm/min	%	8.5
Bending stress	178	2mm/min	MPa	59
Modulus of elasticity in bending	178	2mm/min	MPa	2600
Tensile elasticity modulus	527-1,-2	50mm/min	MPa	2500
Notched Charpy Impact strength	179-1	1eA	kJ/m^2	19
THERMAL				
Vicat softening point	306	50N	°C	88
Deflection temperature under load	75-1,-2	1,8 MPa	°C	71
Coefficient of linear thermal expansion II/⊥	11359-1/-2	23°C - 85°C	E-6/°C	85
COMBUSTIBILITY				
Flammability level test	UL94	1,5 mm	Class	V0
Flammability level test	UL94	3,0 mm	Class	V0
Flammability level test	UL94	6,0 mm	Class	V0

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

There might be some problems to achieve proper adhesion on glass bed.
Using perforated bed, PVA glue, specialized product (Dimafix), kapton coating or other different way to obtain adhesion is recommended.

Be sure to use only with fume extraction or in very well-ventilated conditions!

Recommended parameters of printing:

Hotend temperature	230 - 260 °C
Bed temperature	> 80 °C
Print speed	< 65 mm/s

Examples of problems and their solutions:

Problem	Possible cause	Proposed solution
Weak layers adhesion	1) Extruder temperature is too low 2) Too high speed of printing	1) Raising the extruder temperature 2) Raising the extruder temperature / reducing the print speed
Uneven feeding – losing steps of the feeder / filament sliding on the drive gear	1) Extruder temperature is too low 2) Weak feader pressure	1) Raising the extruder temperature 2) Raising the pressure
Print is peeling of the bed	1) Too low temperature of bed 2) Improperly prepared surface 3) Cooling	1) Raising the bed temperature 2) Degreasing the bed / using another source of adhesion / using perforated bed 3) Giving up the cooling on first stage is recommended
Edge curling	1) Processing shrinkage	1) Compensation with amount of served filament / selection cooling parameters / changing the chamber temperature

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
ABS needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

It's forbidden to set fire or exceed decomposition temperature!

Decomposition of ABS is typically over 300 °C.
Main ingredient of decomposition is styrene. Printing at too high a temperature may cause excessive release of flame retardant vapors.

Detailed safety information available in SDS.